

PREDICTING APARTMENT RENTAL PRICES IN SWITZERLAND

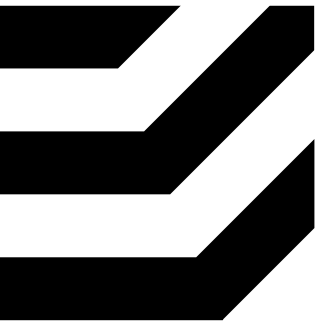
DSPRO1 Project

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AGENDA

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The Problem

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Scope & Tools

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Data Source & Enrichment

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Modelling approach

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Results and analysis

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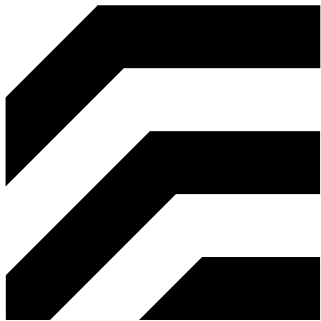
Demo

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Wrap-up

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Questions & Feedback



1

THE PROBLEM

THE PROBLEM

You find a listing
3 rooms, 80 m², Zürich

CHF 2 400 / mo
Is this fair?

Why it's hard to judge

Location dependency
Same flat costs 2× more across cantons

Many interacting features
Area, floor, year built, amenities ...

No transparent benchmark
No public reference model exists

→ **We build a supervised ML model that predicts fair rent from listing data**

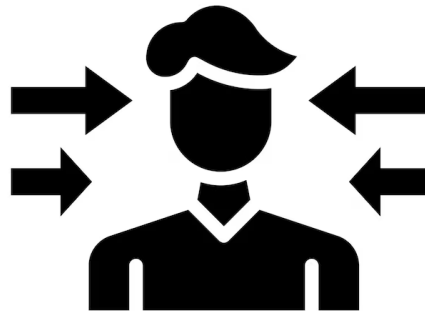
WHY IS IT RELEVANT?



- Tenants



- Landlords



- Socially

HYPOTHESIS

Location-related variables combined with structural apartment features contain sufficient signal to predict monthly rental prices in Switzerland with practically useful accuracy and a Random Forest Regressor will outperform a Linear Regression model in doing so.

2

SCOPE & TOOLS

SCOPE

- **In Scope**

- ✓ Web scraping of Swiss rental listings
- ✓ Data cleaning, deduplication & outlier removal
- ✓ Exploratory Data Analysis (EDA)
- ✓ Feature engineering (location, structure, amenities)
- ✓ Baseline model → Linear Regression → Random Forest
- ✓ Model evaluation: MAE, RMSE, R^2
- ✓ Experiment tracking with MLflow
- ✓ Version control & Kanban via GitHub Projects

- **Out of Scope**

- Cloud deployment or production-ready application
- Real-time or live price feed integration
- Time-series / rental price forecasting
- Non-apartment property types (houses, commercial)
- User-facing UI or web app

PROJECT MANAGEMENT TOOL



- Backlog
- Roadmap
- Milestones

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DATA SOURCE & ENRICHMENT

DATA SOURCE

Comparis

- Lowballed, declined

Immoscout

- Not interested

Rentumo

- Contacted, no answer

Kaggle?

- Not in usable condition

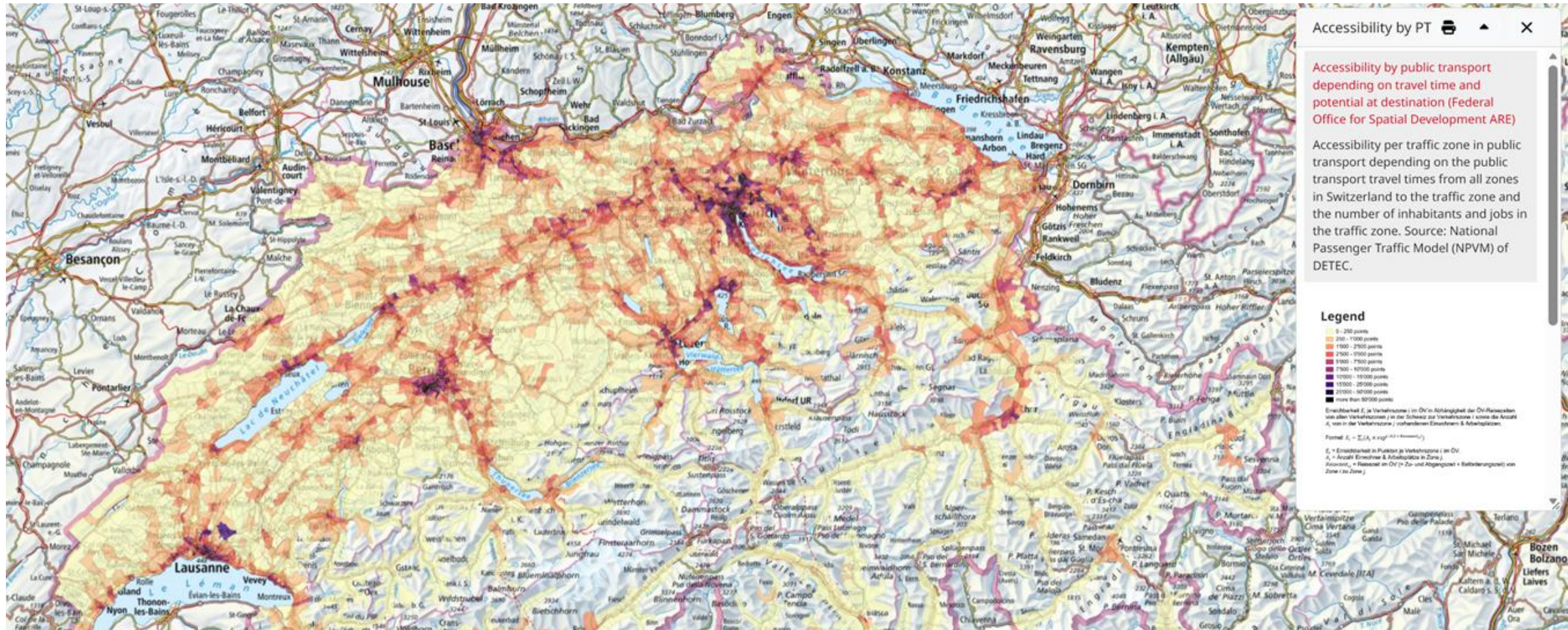
CSV from 2024

- Stale data



DATA ENRICHMENT

- Over API => geo.admin.ch



DATA PIPELINE

1. Scrape listing index

750 pages on rentumo.ch · extracts rooms, area, rent, slug

2. Scrape detail pages

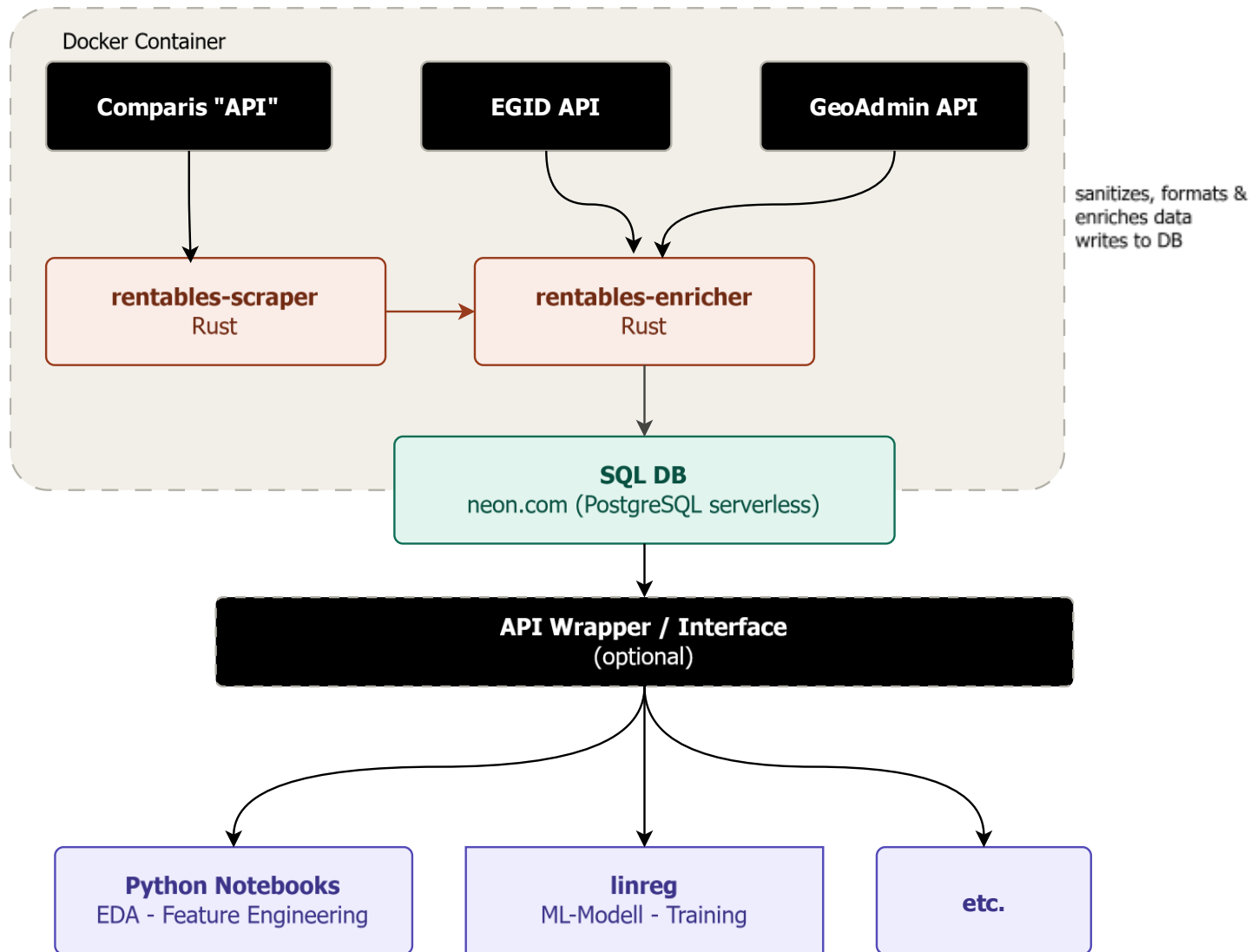
One call per slug · adds address, costs, description, availability

3. Enrich each address

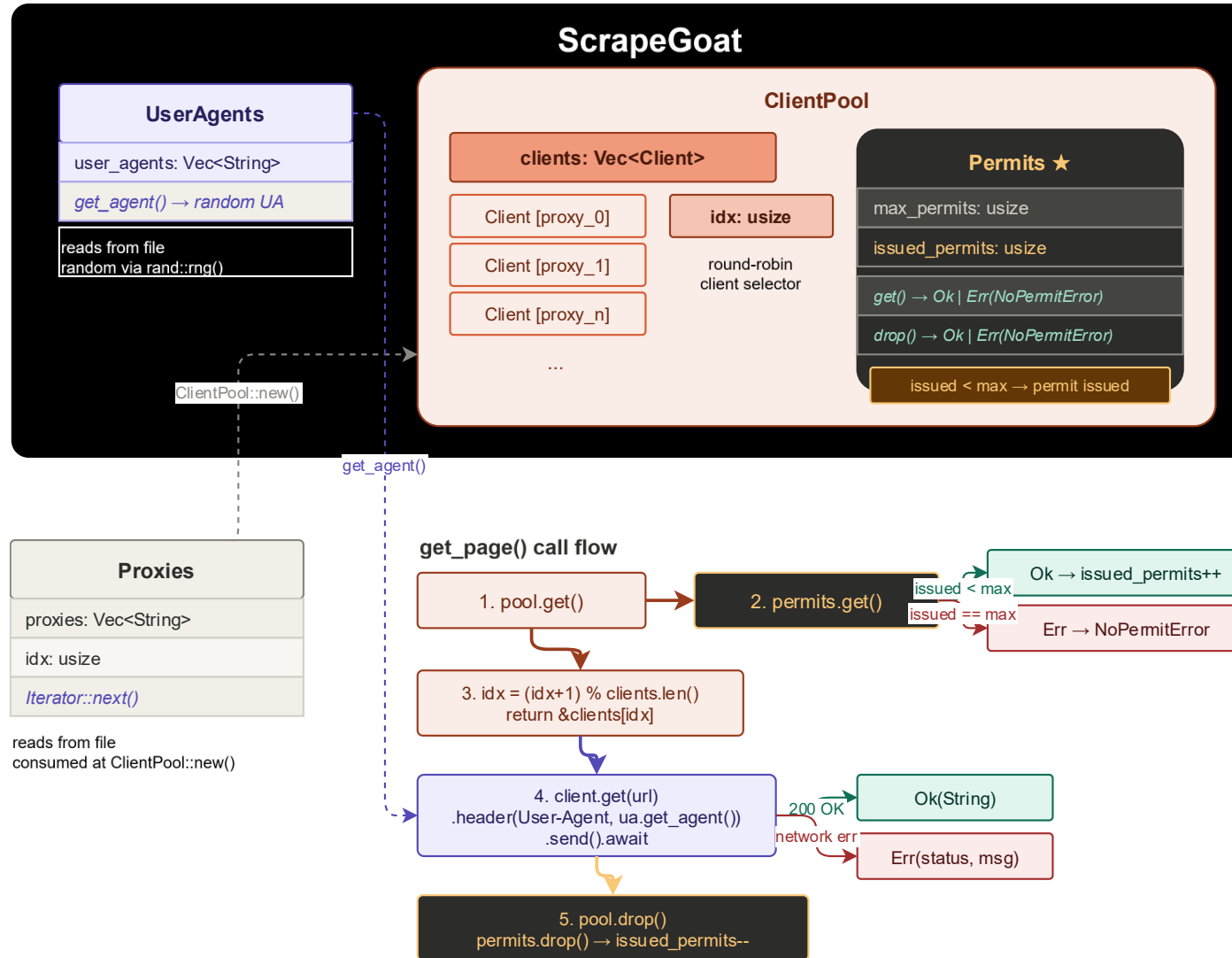
GeoAdmin → coordinates + EGID · BFS GWR → year built, size

swisstopo → elevation, transit score, solar class, population

ARCHITECTURE



POOL & PERMIT SYSTEM



FROM URLS TO A DATASET

25'002 URLs identified on rentumo.ch

↓ 15'008 empty placeholder URLs removed

9'994 listings successfully scraped

↓ Missing address, price, or area

~5'000 after mandatory-field filter

↓ GWR enrichment join + outlier filter

~4'500 final modelling set ← used for training

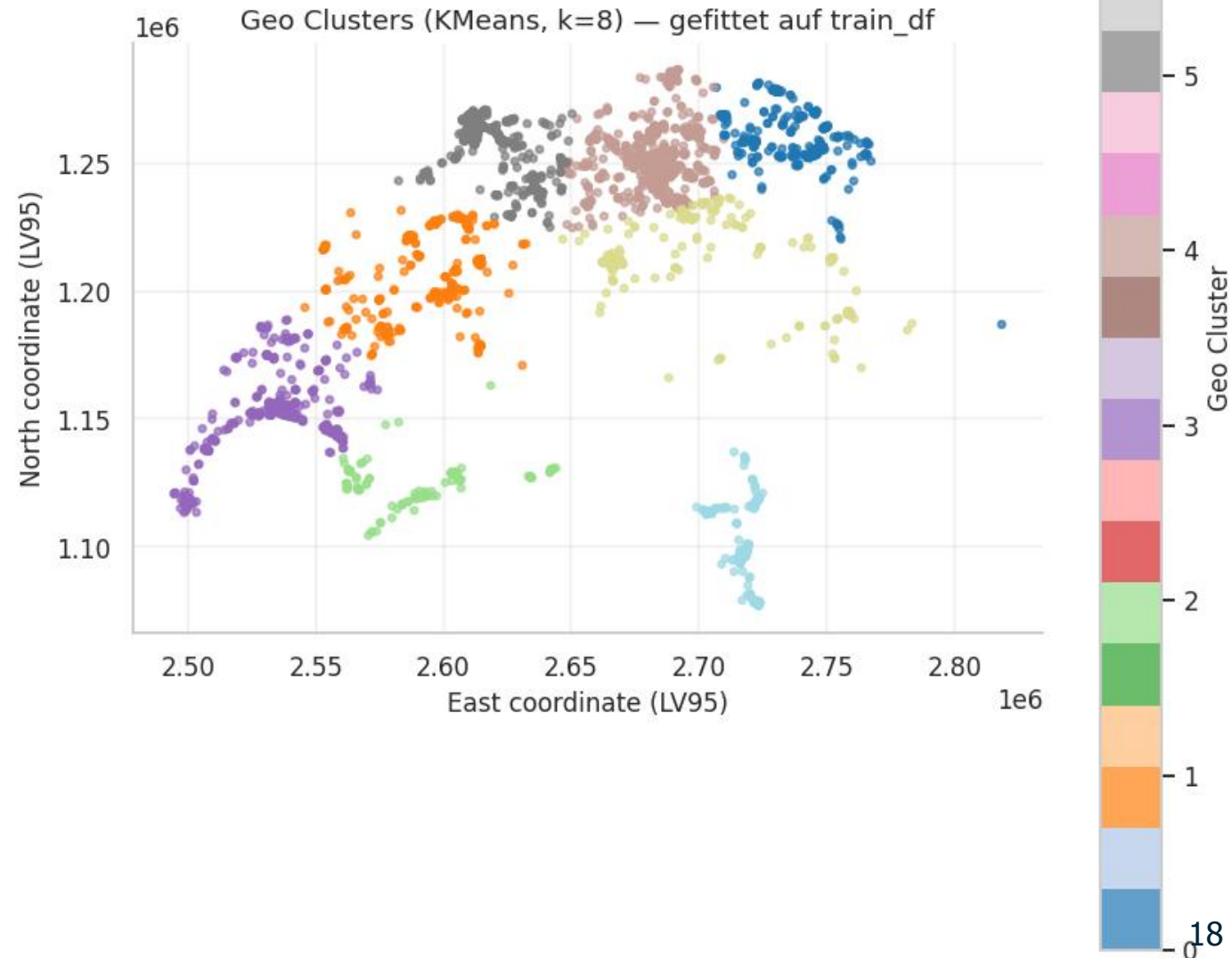
4

MODELLING APPROACH

MODELLING APPROACH

Model pool:

- DummyRegressor (median): lower bound
- Ridge with StandardScaler: linear benchmark
- RandomForestRegressor
- GradientBoostingRegressor
- XGBoost
- LightGBM (best single model)
- StackingRegressor with Ridge meta-learner



EVALUATION PROTOCOL

Hold-out test split (60/20/20)



- **MAE**
Average absolute error in CHF
- **RMSE**
Quadratic error, penalizes large misses
- **R²**
Variance explained by the model

MedAE

Robust median absolute error

CV std

Stability across folds

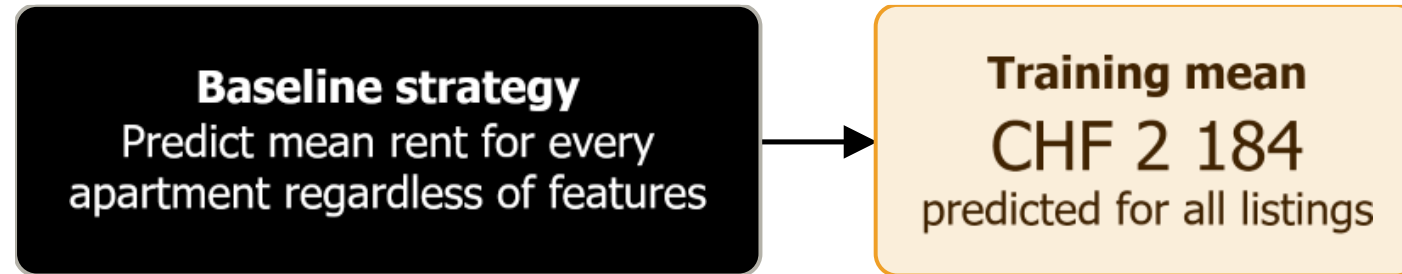
Bootstrap CI

Significance test for model differences

5

RESULTS AND ANALYSIS

BASLINE

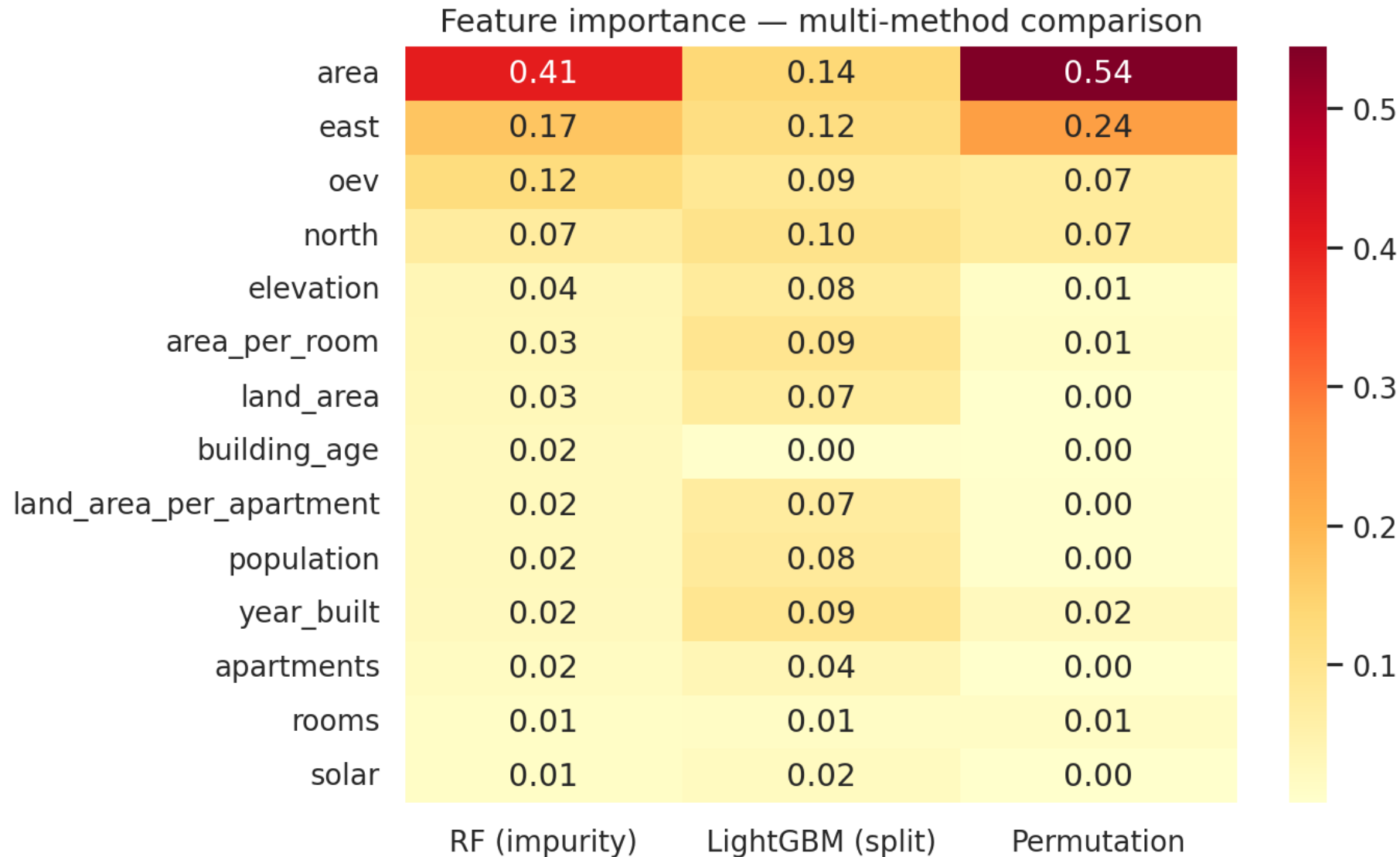


Baseline performance



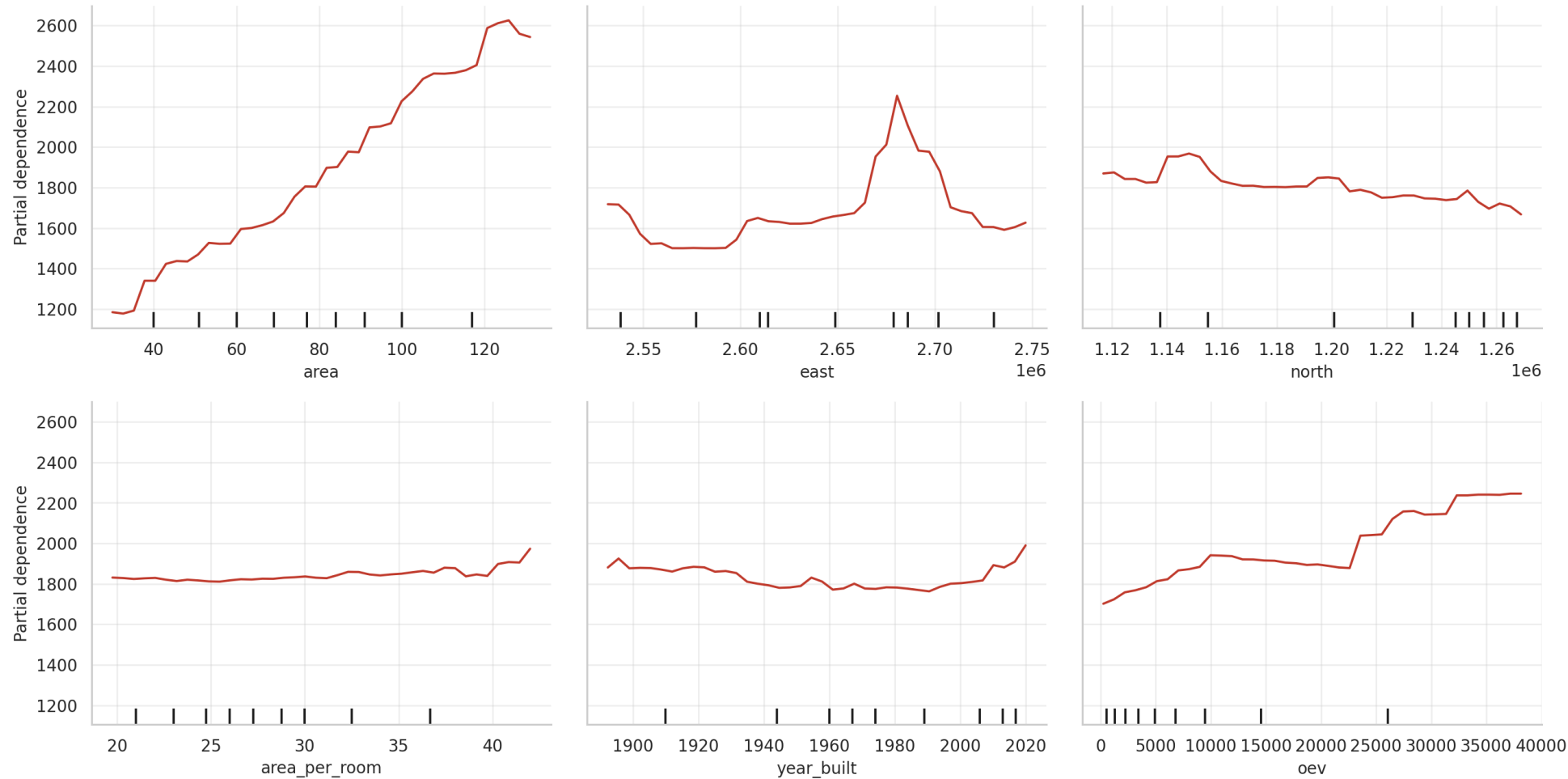
Interpretation
Off by CHF 612 on average — this is our lower bar. Any real model must clearly beat this.

FEATURE IMPORTANCE

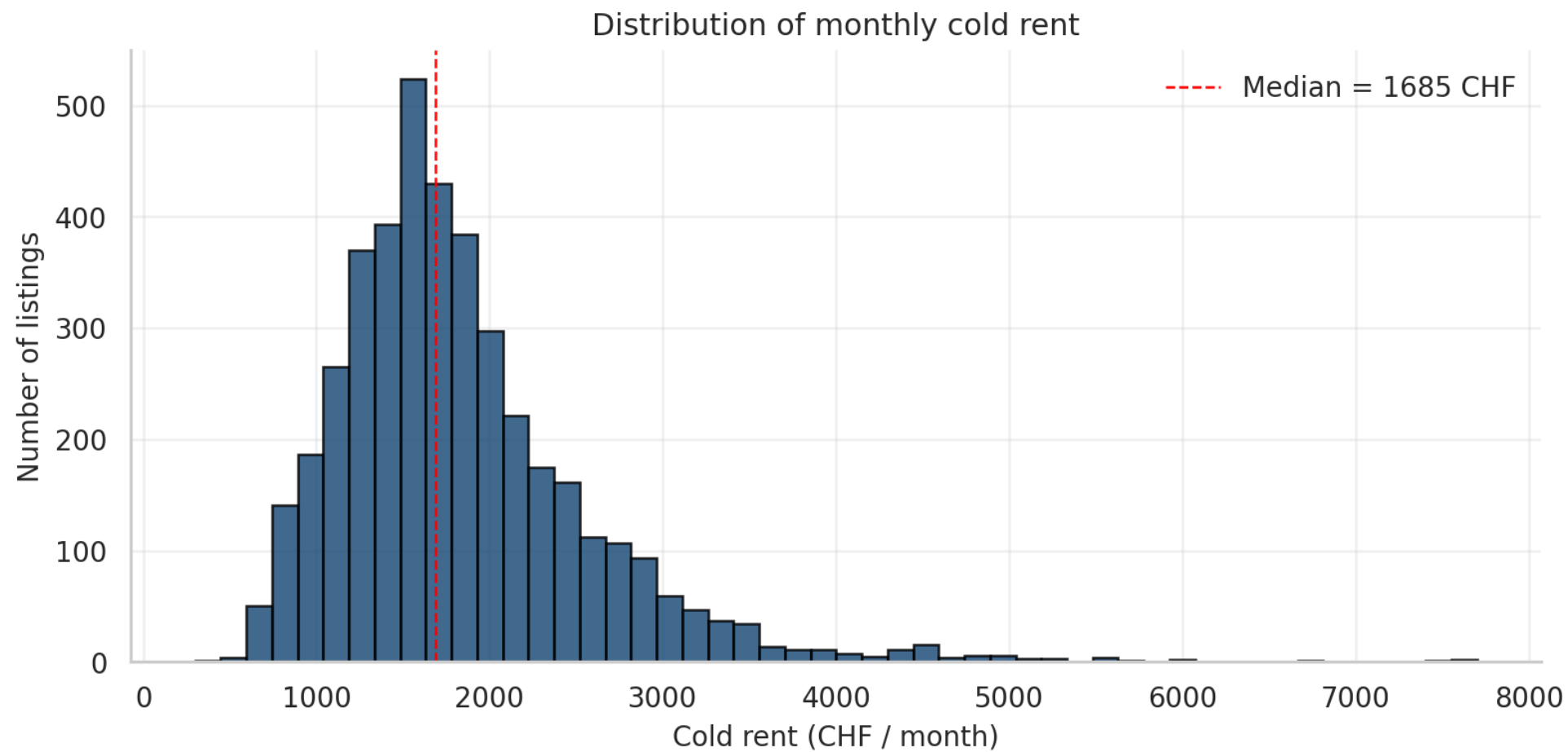


TOP 6 FEATURES

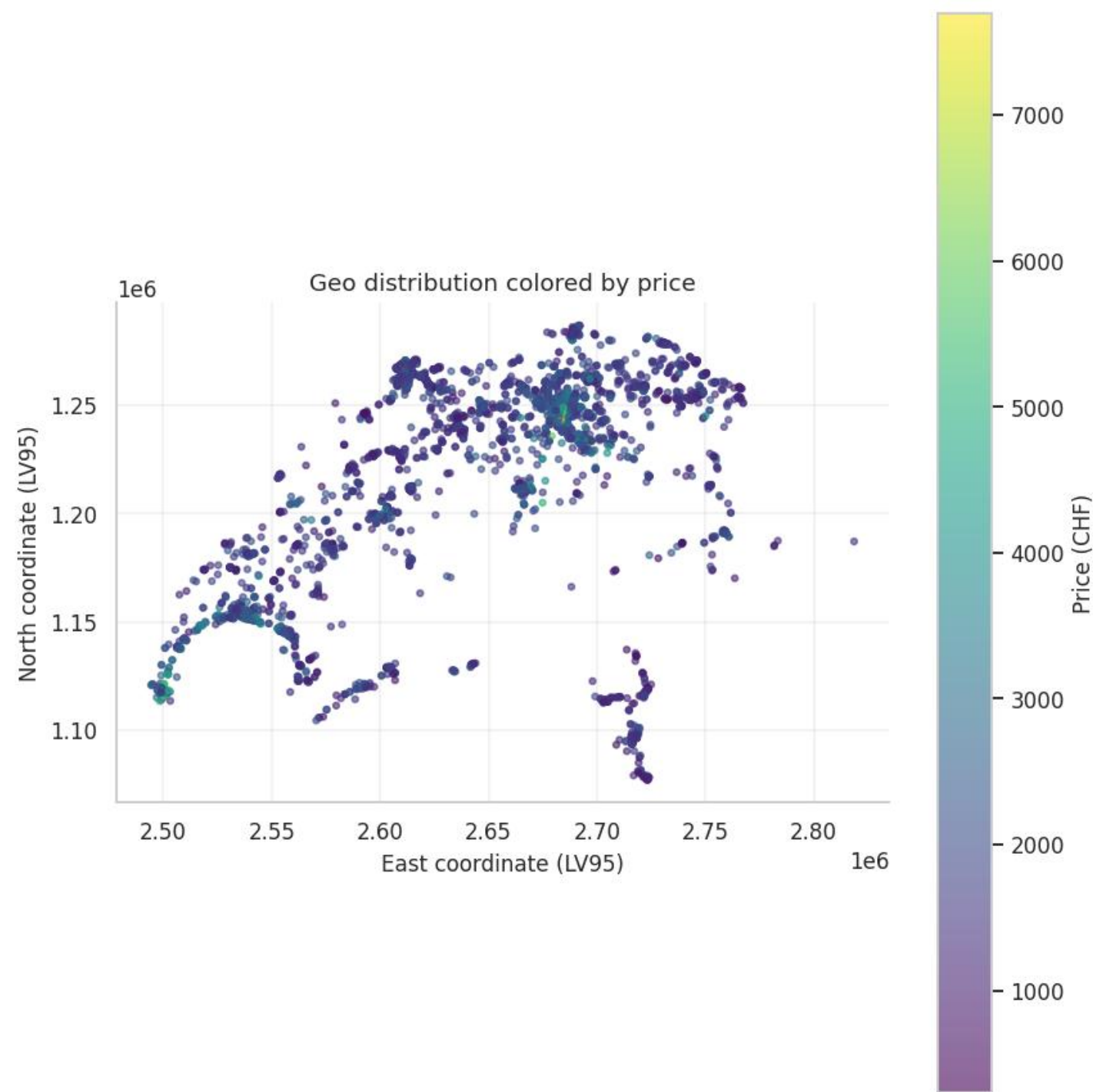
Partial Dependence Plots — Top 6 features (LightGBM)



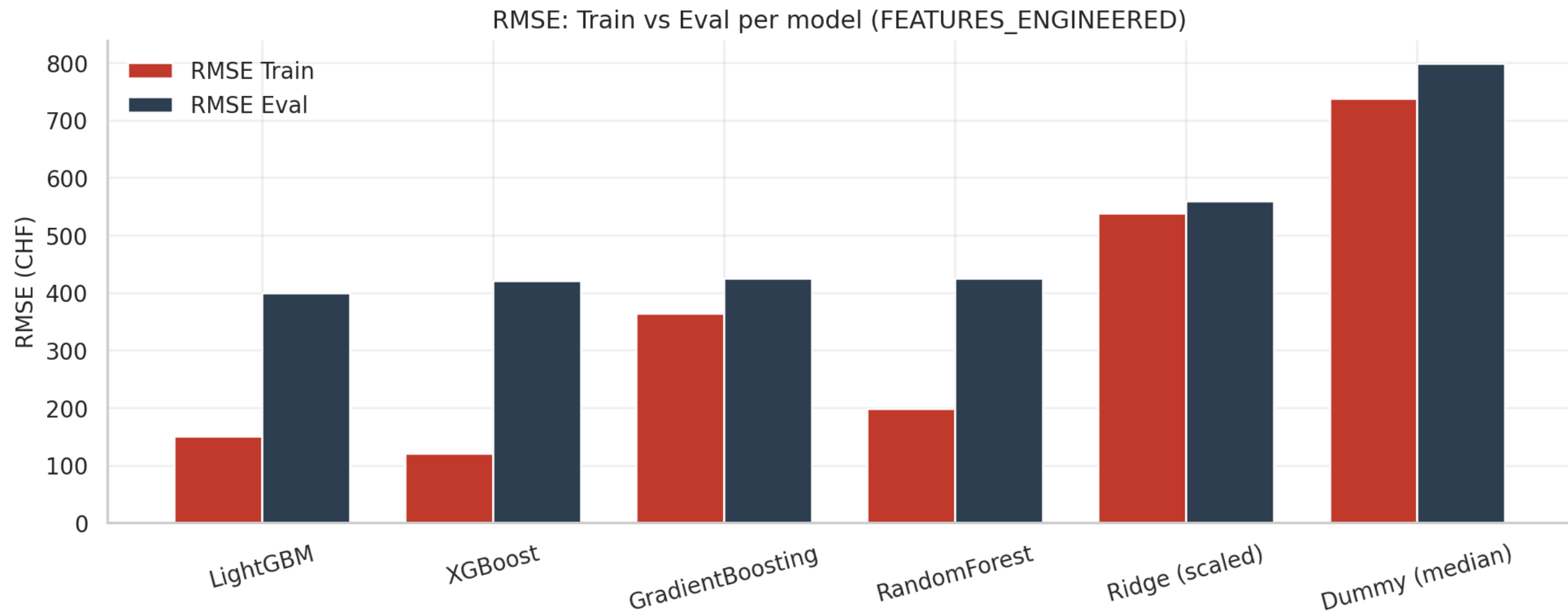
PRICE DISTRIBUTION



GEO PRICE MAP

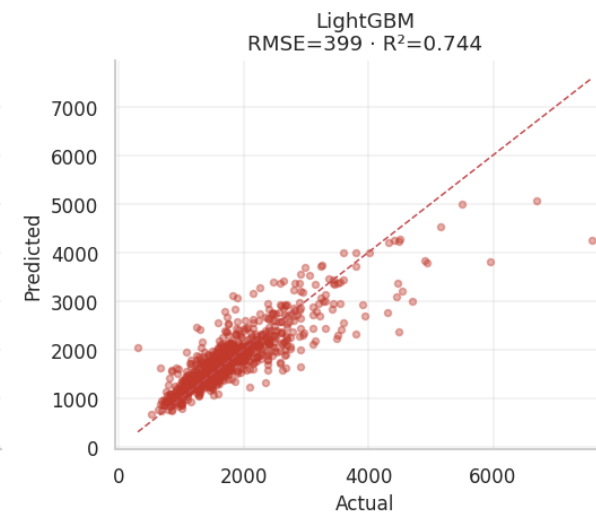
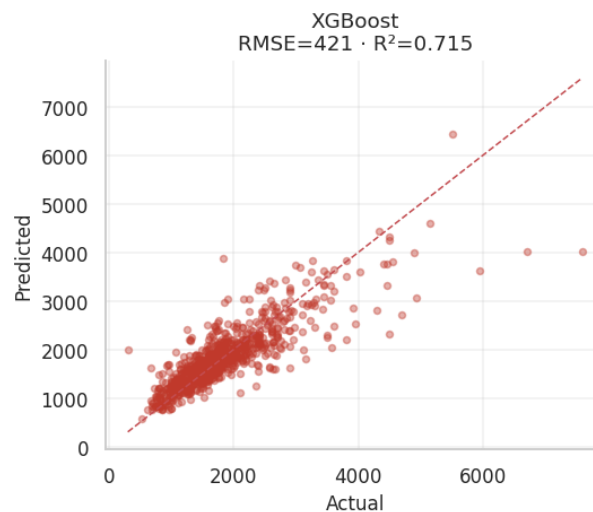
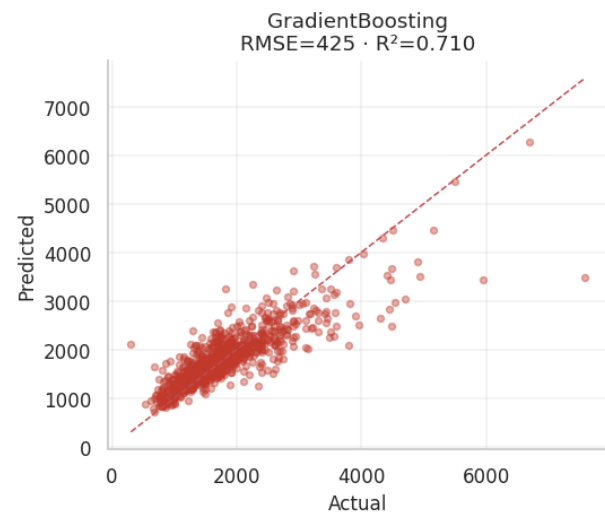
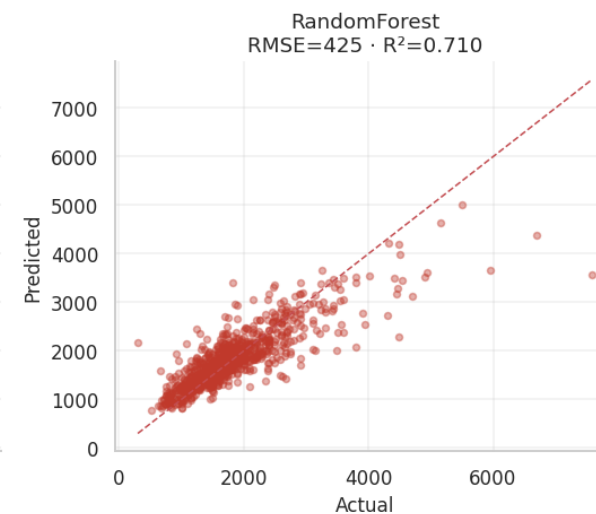
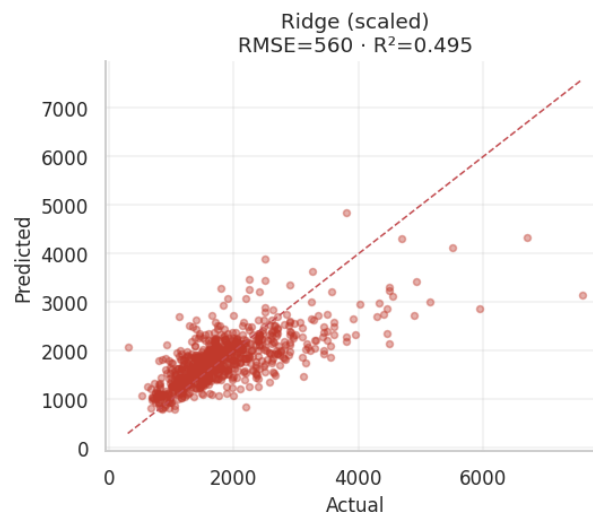
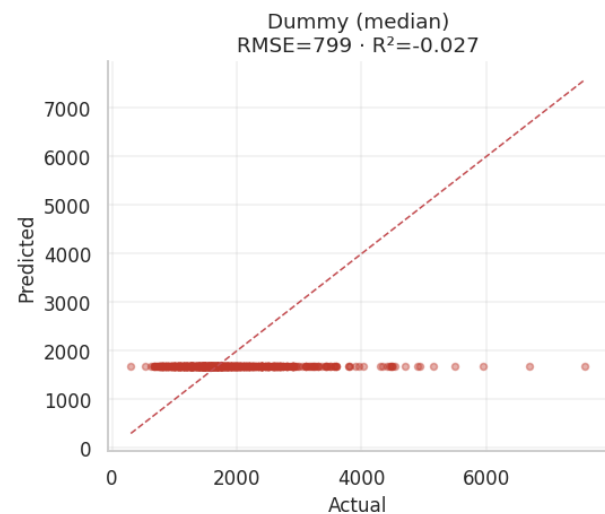


MODEL COMPARISON

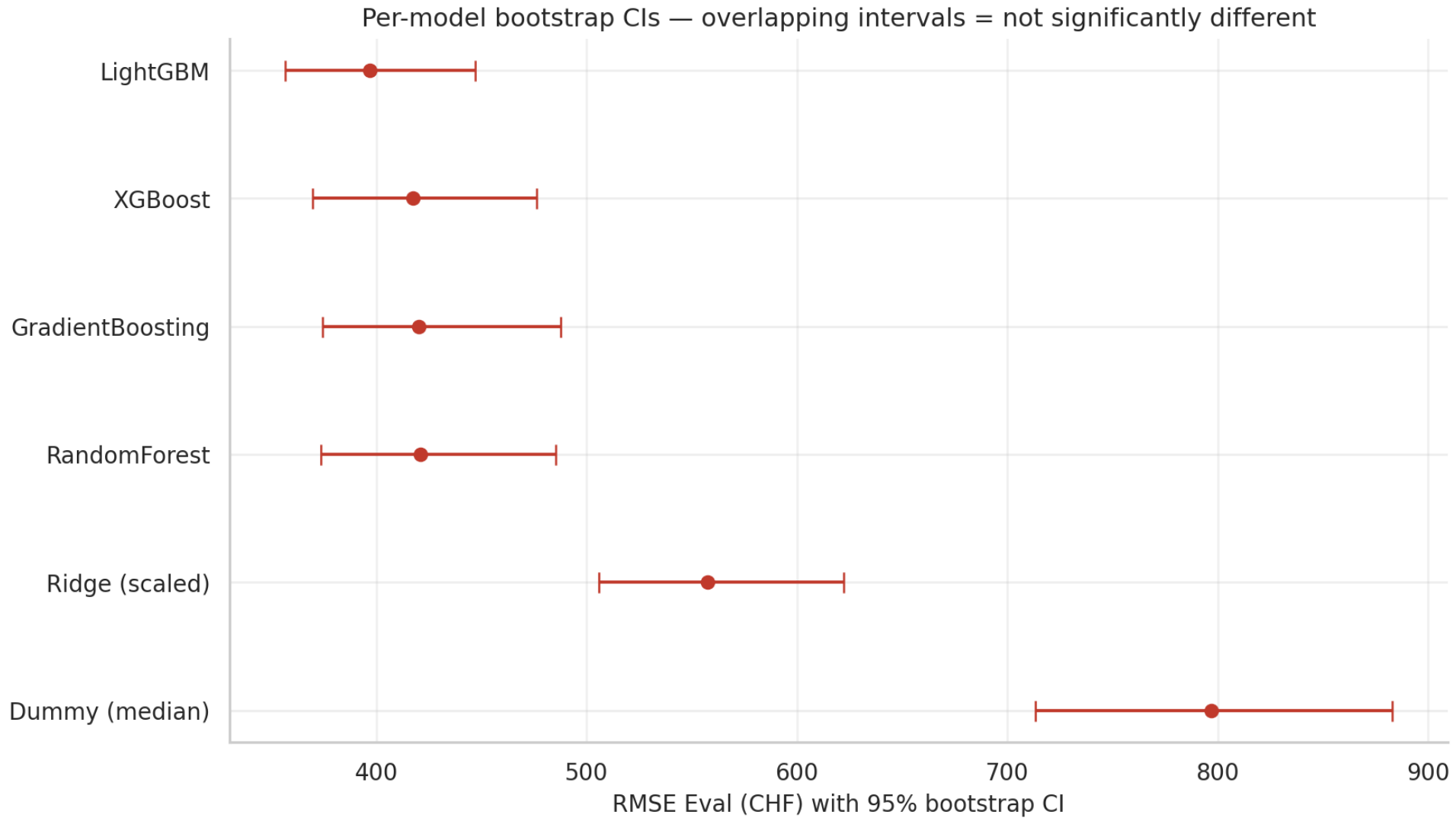


VISUAL MODEL COMPARISON

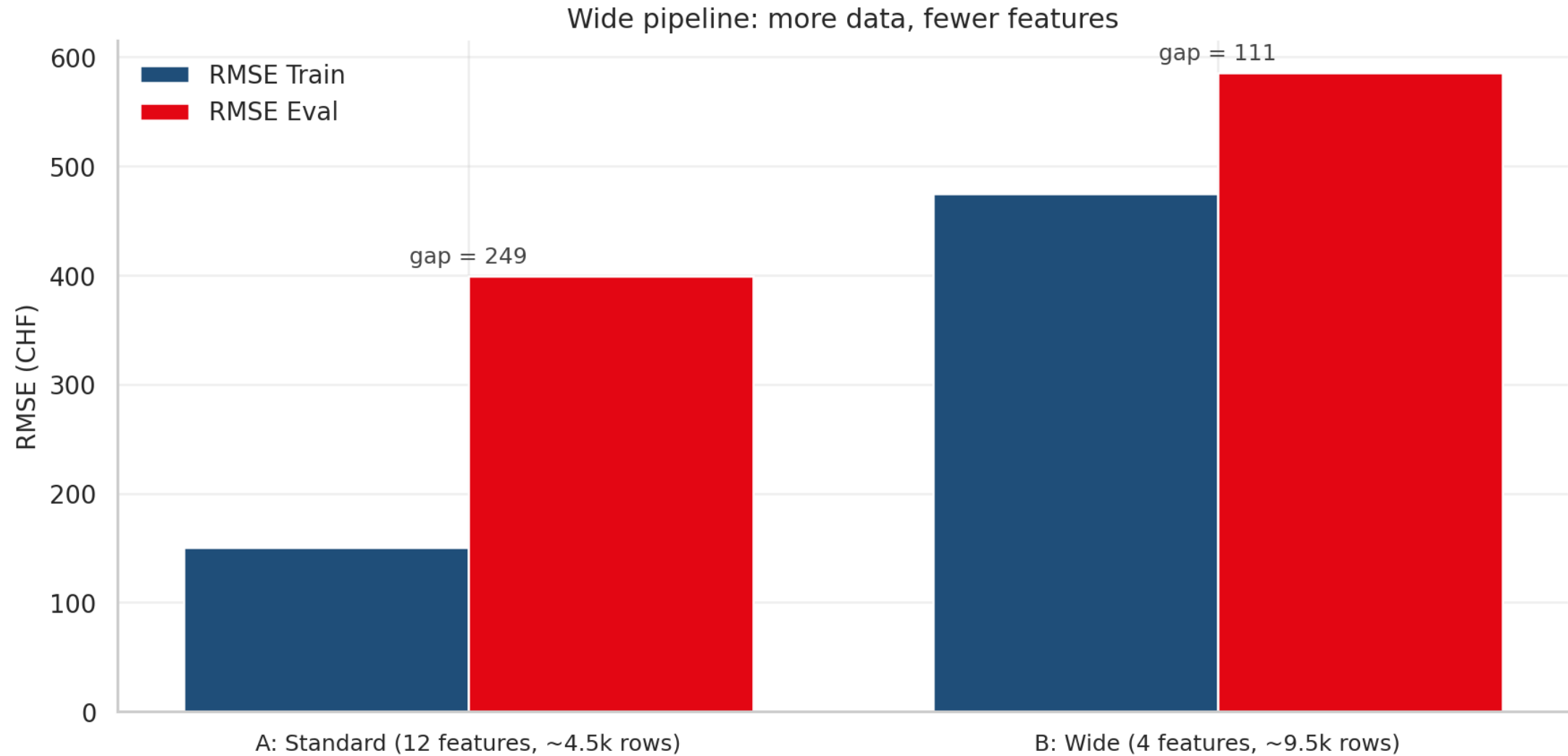
Actual vs Predicted — alle Modelle (Eval-Set)



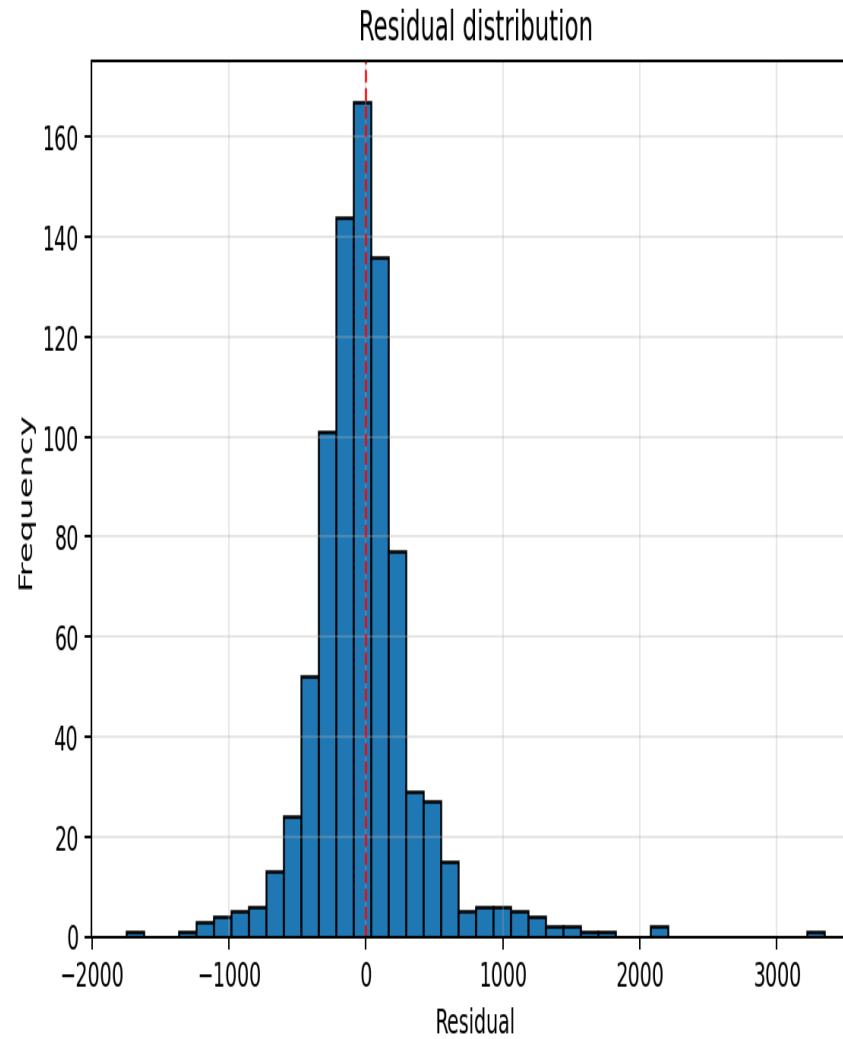
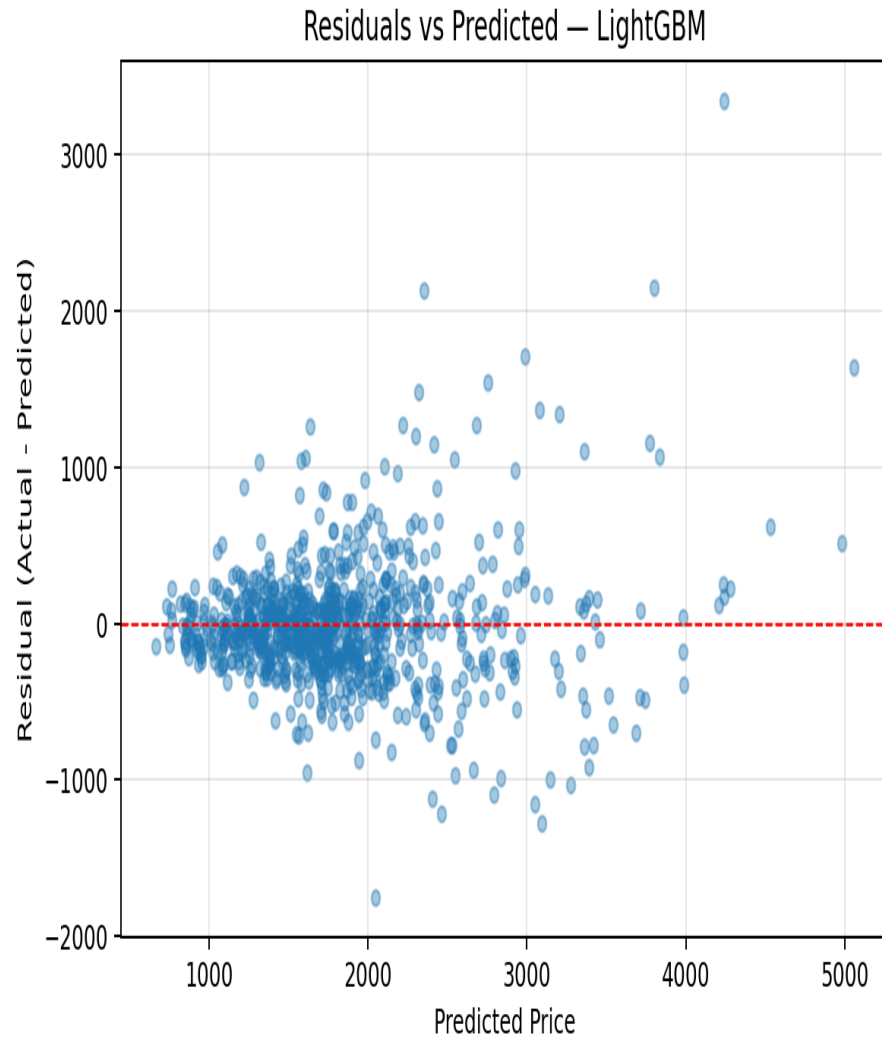
BOOTSTRAPS CLS



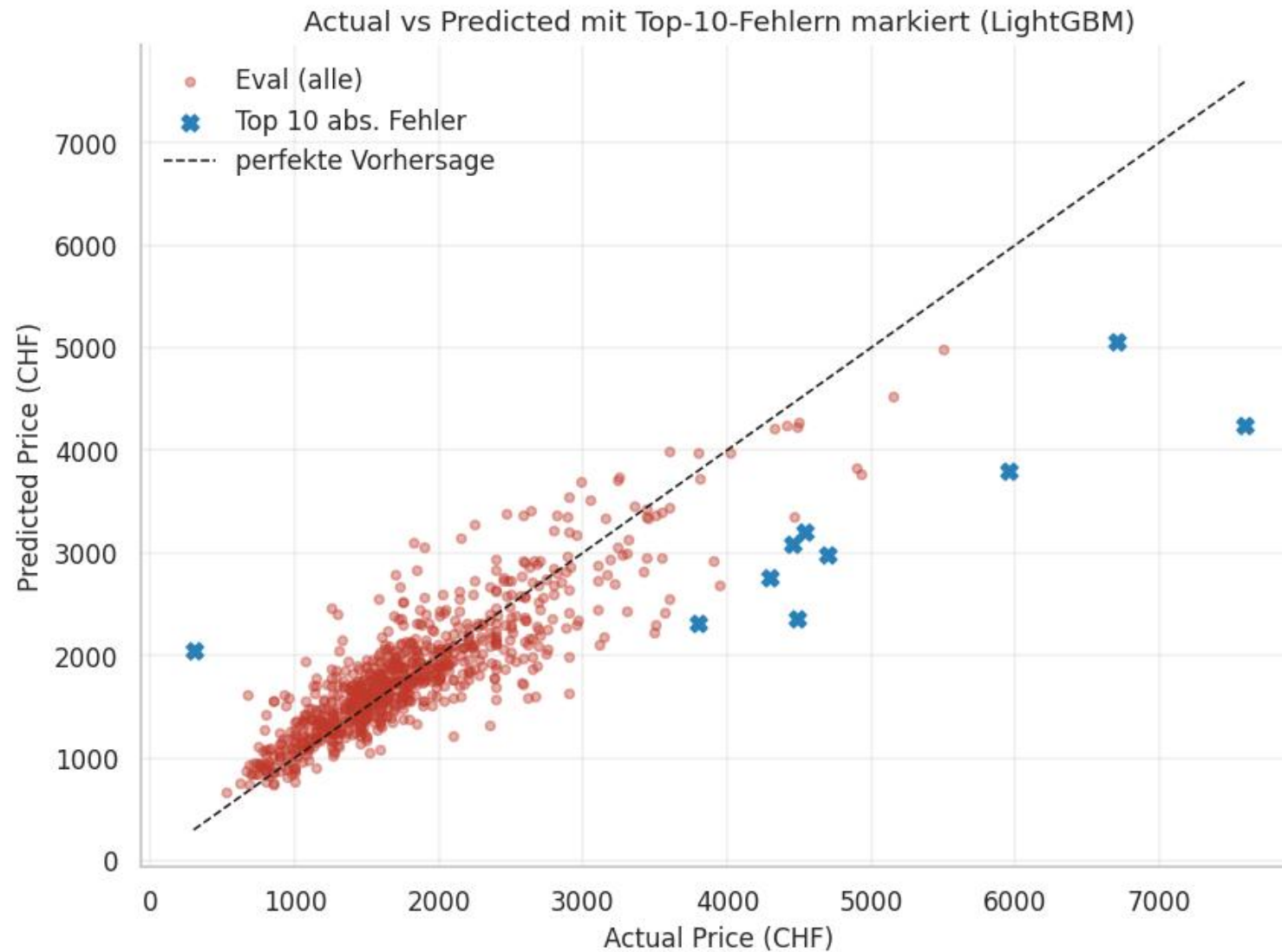
12 VS 4 FEATURES



RESIDUAL ANALYSIS



TOP 10 ERRORS



ERROR PER PRICE BAND



ALL MODEL COMPARISON

	FeatureSet	Model	MAE Train	MAE Eval	RMSE Train	RMSE Eval	R ² Train	R ² Eval	Overfitting Gap	Diagnose	n_geo_clusters	geo_cluster_cols
0	ENGINEERED+geo	LightGBM	104.106127	265.017656	148.632283	395.356652	0.957889	0.748527	246.724369	Possible overfitting	8	east, north
1	ALL+geo	LightGBM	112.919106	264.721002	162.831986	397.848403	0.949458	0.745348	235.016417	Possible overfitting	8	east, north
2	ENGINEERED	LightGBM	104.406305	264.420015	150.286423	399.166141	0.956946	0.743658	248.879718	Possible overfitting	0	
3	ALL	LightGBM	113.217236	266.221322	164.153581	403.349280	0.948634	0.738257	239.195699	Possible overfitting	0	
4	ALL	XGBoost	96.942429	266.688721	139.462017	407.013936	0.962925	0.733479	267.551918	Possible overfitting	0	
5	ALL+geo	RandomForest	124.714897	273.709801	199.692919	414.693447	0.923985	0.723327	215.000529	Possible overfitting	8	east, north
6	ALL+geo	XGBoost	97.636948	269.622589	140.518675	417.280668	0.962361	0.719864	276.761993	Possible overfitting	8	east, north
7	ENGINEERED+geo	XGBoost	85.915131	267.561340	123.294351	418.444105	0.971022	0.718300	295.149753	Possible overfitting	8	east, north
8	ENGINEERED+geo	RandomForest	123.804368	275.576335	197.556061	419.068587	0.925603	0.717458	221.512526	Possible overfitting	8	east, north
9	ALL	RandomForest	125.550257	274.519146	201.437937	419.549506	0.922651	0.716809	218.111570	Possible overfitting	0	
10	ENGINEERED	XGBoost	84.069862	269.927094	121.598850	420.806889	0.971814	0.715109	299.208039	Possible overfitting	0	
11	ALL+geo	GradientBoosting	254.245868	281.249725	366.737568	423.397067	0.743619	0.711591	56.659499	Good generalization	8	east, north
12	ENGINEERED	GradientBoosting	254.019719	281.771860	364.777231	424.709516	0.746353	0.709801	59.932284	Good generalization	0	
13	ENGINEERED	RandomForest	124.279534	276.874338	198.916167	424.910709	0.924575	0.709526	225.994542	Possible overfitting	0	
14	ENGINEERED+geo	GradientBoosting	253.913457	281.643481	362.887421	425.289407	0.748974	0.709008	62.401986	Good generalization	8	east, north

HYPOTHESIS CONFIRMED

- **Midterm baseline**
- CHF 847
- RMSE on evaluation set
- $R^2 = 0.00$
- (No variance explained)



GradientBoosting + KNN features

CHF 423 RMSE on evaluation
set

$R^2 = 0.71$

6

DEMO

DEMO: STREAMLIT APP

Mietpreis-Schätzer Schweiz - Google Chrome

localhost:8501

New Chrome available

Deploy

Modell auswählen

best_model_v3.joblib · Defa...

Modell aktiv:
best_model_v3.joblib ·
LGBMRegressor

Dashboard v4.3 · Script OK

Manuelle Schätzung

Wohnfläche (m²)

75

Zimmer

3

Baujahr

1990

Gebäude

Wohnungen im Gebäude

8

Grundstücksfläche (m²)

300

Mietpreis-Schätzer Schweiz

Adresse → EGID/GWR → Swisstopo → Modellvergleich → transparente Kaltmiet-Schätzung

Version: Dashboard v4.3

☒ Schätzung ☐ Model Performance ☐ Modell & Daten

Schätzung

Adresse auswählen, Wohnung wählen, Werte prüfen und Kaltmiete schätzen.

Offizielle Adresse suchen

Tippe Strasse/Hausnummer. Ort oder PLZ ist optional; der Vorschlag kommt von GeoAdmin.

Adresse suchen

z. B. Kirchhaldenstrasse 36b oder Kronenbergstrasse 5

- Kronenbergstrasse 3 8800 Thalwil
- Kronenbergstrasse 4 8800 Thalwil
- Kronenbergstrasse 5a 8800 Thalwil
- Kronenbergstrasse 5 8800 Thalwil
- Kronenbergstrasse 7 8800 Thalwil
- Kronenbergstrasse 9 8800 Thalwil
- Kronenbergstrasse 11 8800 Thalwil

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WRAP-UP

LIMITATIONS

Snapshot dataset

One scrape on 13 April 2026; market shifts not reflected

Urban bias

Zurich, Geneva, Basel
overrepresented; rural cantons thin

Expensive apartments

Top-quartile MAE is roughly twice the rest. Luxury attributes (view, floor, finish) missing from features

NEXT STEPS

1. Text features from descriptions

Extract balcony, parking, floor, view - biggest lever for expensive band

2. Second rental platform

Add homegate.ch or immoscout24 - reduces urban bias, doubles training data

3. Prediction intervals

Move from point estimates to CHF $X \pm Y$ via conformal prediction (MAPIE)

4. Lightweight hosting

Deploy Streamlit on Streamlit Cloud or Render - first DSPRO2 deliverable

TL:DR

RMSE reduction vs. baseline

> 50%

Tree-based models win clearly

Best model RMSE

CHF 423

GradientBoosting + KNN features

R^2 on held-out set

0.71

Hypothesis confirmed

THANK YOU FOR YOUR ATTENTION.

We are happy to answer any questions or discuss the presented topics further.

HSLU Hochschule
Luzern

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